RESEARCH PROJECT · 01 NOVEMBER 2026

Modeling Human Activity States Using Hidden Markov Models

Smartphone accelerometer and gyroscope signals · Four-state Gaussian HMM · Session-held-out evaluation

SECTION 1

Background and motivation

This study examines whether smartphone inertial sensor streams can distinguish **Standing, Walking, Jumping, and Still**. Such recognition could support basic motion monitoring and activity summaries without camera-based observation. A Hidden Markov Model (HMM) represents underlying activity as hidden states and uses noisy sensor-derived observations to estimate activity sequences.

SECTION 2

Data collection and preprocessing

The dataset contains **52 independent Sensor Logger recordings**, with 13 labeled recordings per activity. Every session includes timestamped triaxial accelerometer and gyroscope measurements. Recorded sampling rates cluster near 99.45 Hz; recordings are synchronized and interpolated onto a common **50 Hz** grid. One-second windows (50 points), advanced in 0.5-second steps, yield consistent time-domain and FFT features.

Activity	Sessions	Recorded duration	90-second threshold
Standing	13	89.204 s	Short by 0.796 s
Walking	13	90.797 s	Met
Jumping	13	91.355 s	Met
Still	13	103.907 s	Met

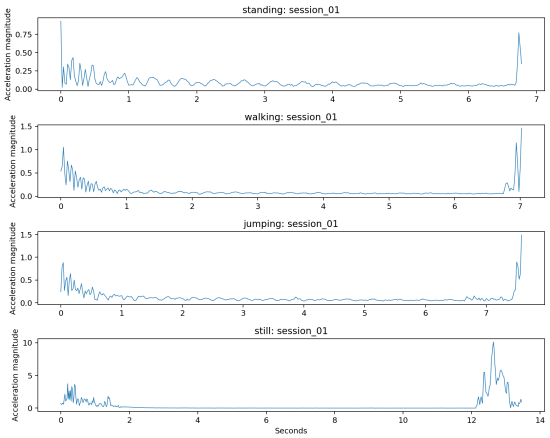


Figure 1. Representative accelerometer-magnitude signals from the four labeled activities.

SECTION 2

Preprocessing and feature engineering

Each archive is treated as a distinct recording session. Sensor timestamps are sorted and duplicate timestamps removed. Accelerometer and gyroscope channels are aligned on a shared resampled time grid. Complete recording sessions are split into training and test groups **before** feature normalization to prevent leakage between overlapping windows.

A fixed 50 Hz output rate enables one-second windows with an FFT frequency resolution of approximately 1 Hz. Rates and durations for every recording are documented in the accompanying dataset audit. These recordings were collected from the supplied phone data; the pipeline is capable of harmonizing different sampling rates, but this study did not test multiple phone models.

Selected observation features

Domain	Features	Purpose
Time	Acceleration standard deviation; dynamic acceleration RMS; dynamic SMA	Motion intensity and variation
Time	Gyroscope magnitude mean and standard deviation	Rotation and stability
Frequency	Dominant nonzero acceleration frequency; acceleration spectral energy; gyroscope spectral energy	Periodic motion and oscillatory intensity

Frequency features are computed from real FFT coefficients after Hann-window tapering and removal of the DC component. Every feature is standardized with a z-score transformation using only the **training-set mean and standard deviation**; these same parameters are applied unchanged to test observations.

Windowing logic

At 50 measurements per second, a 1.0-second window contains 50 observations. This permits estimation of short-term signal variability and frequency content while retaining enough windows per 6–14-second session for sequential decoding. A 0.5-second hop increases temporal resolution, but overlapping windows within a session must never be assigned to different dataset splits.

SECTION 3

HMM design and implementation

The four latent states are **Z = {Standing, Walking, Jumping, Still}**. Each observation vector contains eight standardized signal features. The initial-state distribution π models the starting state; the **4 × 4 transition matrix A** models the probability of successive states; and the emission model **B** is a continuous diagonal-covariance multivariate Gaussian density associated with each state.

Baum–Welch training and Viterbi decoding

Baum–Welch expectation-maximization estimates initial, transition, and emission parameters using a numerically stable log-space forward–backward algorithm. Convergence is assessed using the relative change in log-likelihood; a 70-iteration maximum is only a safety cap. The recorded optimization converged in **39 iterations**. Viterbi decoding returns the most probable state sequence. The mapping of latent clusters to activity names is estimated from training recordings, not from the held-out test labels.

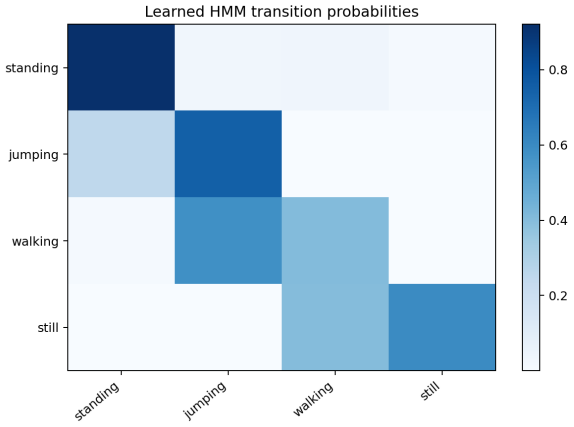


Figure 2. Learned HMM state-transition matrix after training-based label alignment.

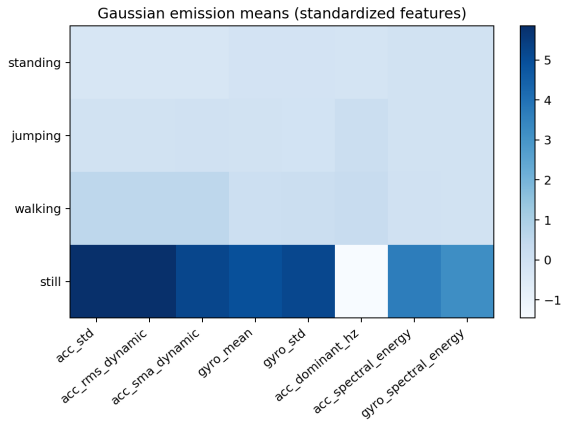


Figure 3. Learned Gaussian emission means expressed in standardized feature units.

SECTION 4

Evaluation on unseen sessions

The held-out evaluation uses **eight complete recording sessions** (two per class); the other 44 sessions are used for training. This session-level separation prevents related overlapping windows from appearing in training and testing. Accuracy across the **101 withheld windows** was **21.78%**, indicating inadequate recognition performance.

Activity	Test windows	Sensitivity	Specificity	1-vs-rest accuracy
Standing	25	0.400	0.500	0.475
Walking	25	0.120	0.895	0.703
Jumping	24	0.375	0.571	0.525
Still	27	0.000	1.000	0.733

Sensitivity = TP/(TP + FN); specificity = TN/(TN + FP); and one-vs-rest accuracy = (TP + TN)/N. These are class-wise binary metrics, distinct from the overall multiclass accuracy. The model correctly identified no Still windows in this particular held-out set.

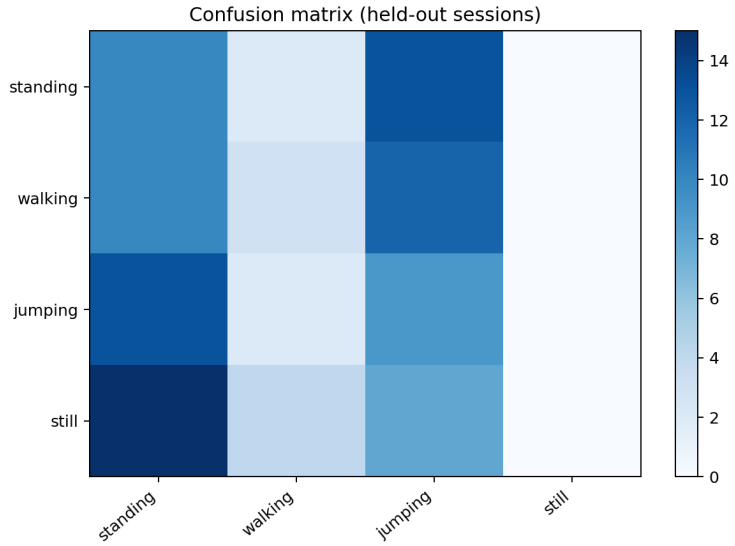


Figure 4. Confusion matrix for withheld test sessions.

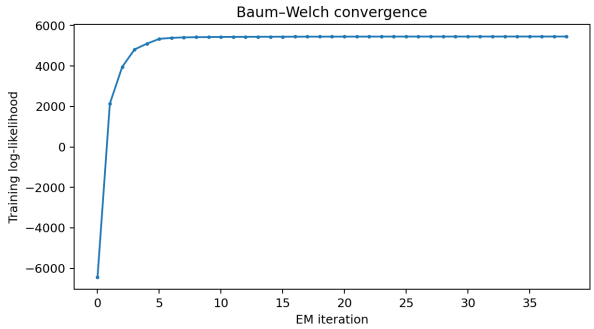


Figure 5. Baum–Welch training log-likelihood trajectory.

SECTION 5

Results interpretation and reflection

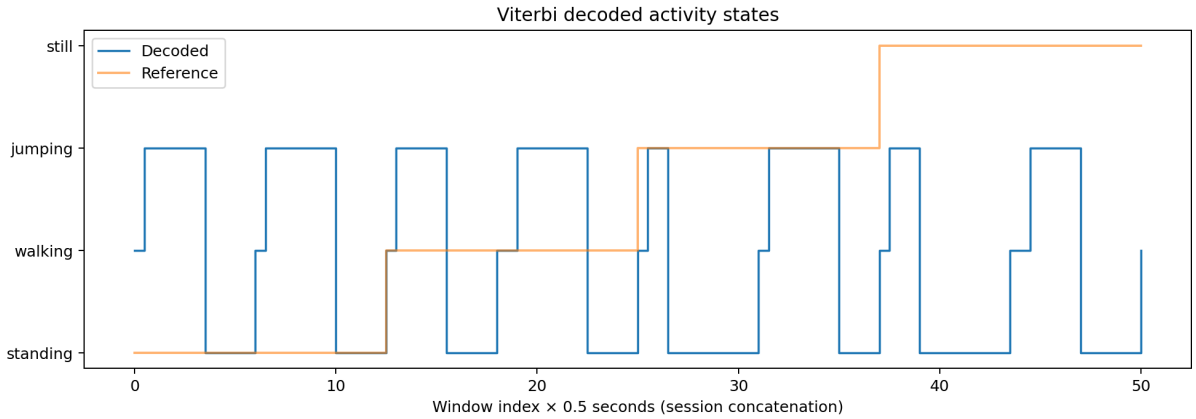


Figure 6. Viterbi predictions versus recording labels across the held-out sessions.

Standing was the least poorly recovered class (sensitivity 0.400), while Still had zero recall and Walking achieved only 0.120. The learned transition probabilities describe inferred state persistence **within** single-activity clips; they should not be interpreted as verified human transitions between activities. Signal noise, phone orientation, vibration, and placement can substantially affect both time- and frequency-domain features.

SECTION 6

Discussion and conclusion

The complete computational workflow—data ingestion, timestamp alignment, rate harmonization, windowed feature extraction, training-only standardization, Baum–Welch parameter estimation, Viterbi decoding, and session-held-out evaluation—was implemented and reproduced. **However, the 21.78% test accuracy does not support a reliable activity-recognition claim.** Some Standing, Walking, or Jumping clips were reportedly recorded with a stationary tripod, in which case the sensors measure the phone rather than the participant's body. More reliable research requires carrying or attaching the phone consistently during body movement, collecting independent trials, and testing genuine transitions.

Recording evidence and submission details

The assignment requires four distinct, well-lit activity-verification videos (at least five seconds each), showing the participant's face, full body and recording process. Video URLs and the GitHub repository URL must be added to the submitted report. The provided data include 52 sessions, but Standing totals 89.204 seconds, which is 0.796 seconds below the stated minimum.

Video links: Standing [insert URL] | Walking [insert URL] | Jumping [insert URL] | Still [insert URL]

Repository: [insert GitHub URL]